

User Interaction and Usability of Digital Products

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Human-Computer Interaction

CLASS-13

Visualising Your Product

Visualising your product refers to the process of creating a clear, tangible, and often graphical representation of your product idea to convey its functionality, design, and value to users, stakeholders, and team members.

It helps bridge the gap between an abstract concept and a concrete solution by enabling people to understand, interact with, and provide feedback on the product before it is fully developed.

Interaction Design Between User & Features

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Usability is a **quality attribute** that assesses how easy user interfaces are to use. The word "usability" also refers to methods for improving ease-of-use during the design process.

Usability is defined by **5 quality components**:

- **Learnability:** How easy is it for users to accomplish basic tasks the first time they encounter the design?
- **Efficiency:** Once users have learned the design, how quickly can they perform tasks?
- **Memorability:** When users return to the design after a period of not using it, how easily can they reestablish proficiency?
- **Errors:** How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
- **Satisfaction:** How pleasant is it to use the design?

User Task Analysis & Task Modelling

Listing of actions a user carries out
in performing a task

Task Analysis

One of the most important steps in the design thinking or human-centered design (HCD) methodology is to define the users' problems.

Task analysis is a method that helps you understand how users accomplish their goals and the steps they take to get there. The most common output of a task analysis is a diagram that outlines the user's actions and the system's responses.

This means to clearly identify and articulate problems in the User Interaction and User Experience so that you can later begin the ideation process (i.e., generate great ideas on how to solve said problems).

Task Analysis

Example: *a person goes to a restaurant to eat food :*

- 1. Reach restaurant*
- 2. Browse menu*
- 3. Order Food*
- 4. Wait for food to be served*
- 5. Verify if the ordered food is served*
- 6. Eat ordered food*
- 7. Pay the bill*
- 8. Leave Restaurant*

In Human Computer Interaction, task analysis is the recording of physical and perceptual actions of the user whilst executing the task.

Task Analysis

Designers/Engineers can use Task Analysis to identify areas where additional support may be needed or to eliminate unnecessary steps. For example, designers may automate certain actions that users currently perform manually.

One can approach a task analysis activity from two main viewpoints:

Hierarchical Task Analysis: Consists of breaking down tasks into smaller sub-tasks.

Cognitive Task Analysis: Focuses on tasks that require decision-making, problem-solving, memory, attention and judgment.

Task Analysis

To ensure an effective task analysis process, it's essential to gather focused data during user research.

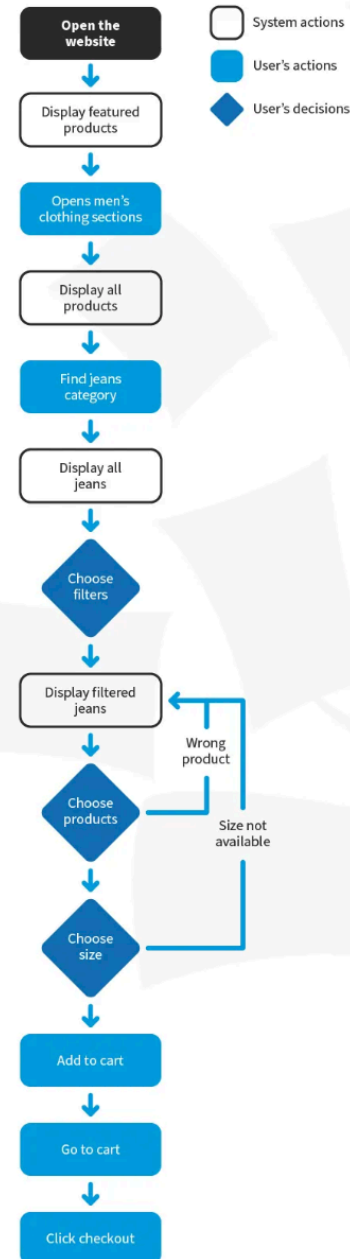
Cognitive scientist and UX consultant Larry Marine suggests collecting five types of data during this phase:

- **Trigger:** What initiates the task for the user?
- **Desired Outcome:** How will users know they have successfully completed the task?
- **Base Knowledge:** What will the users be expected to know when starting the task?
- **Artefacts:** What resources or tools will users require during the task?

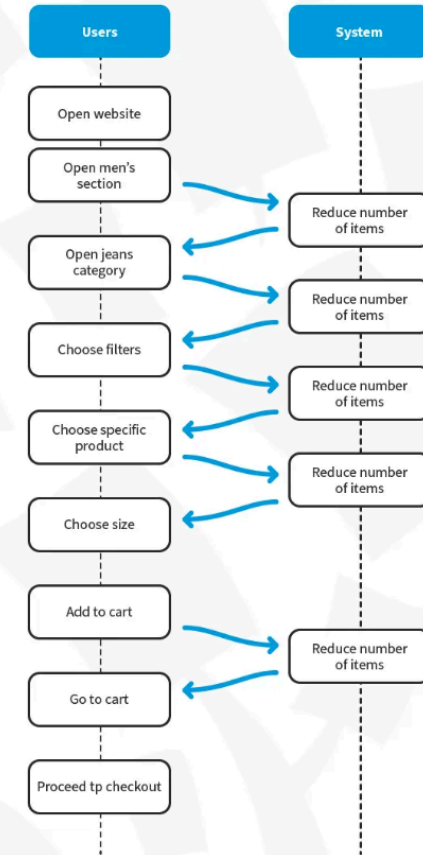
Task Analysis

With your current information (data collected from user research) , you can sketch out how a user goes about their daily life by mapping out the sequence of activities required to achieve a goal.

Flow Diagram



Sequence Diagram



User Task Analysis and Modelling

Listing of actions a user carries out
in performing a task

Task Modeling

Distribution of Tasks between
Users and Technology (Computer)

Task Analysis

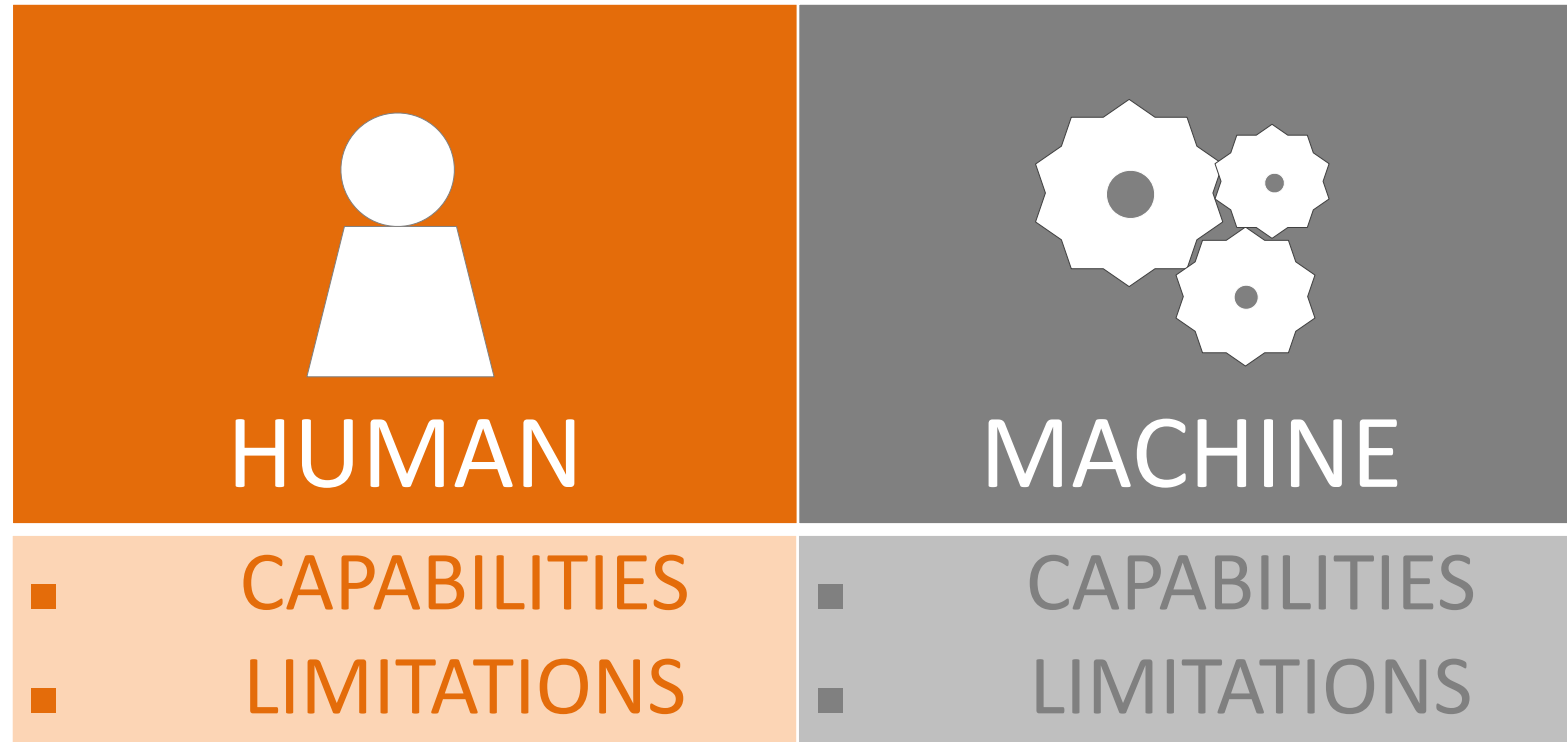
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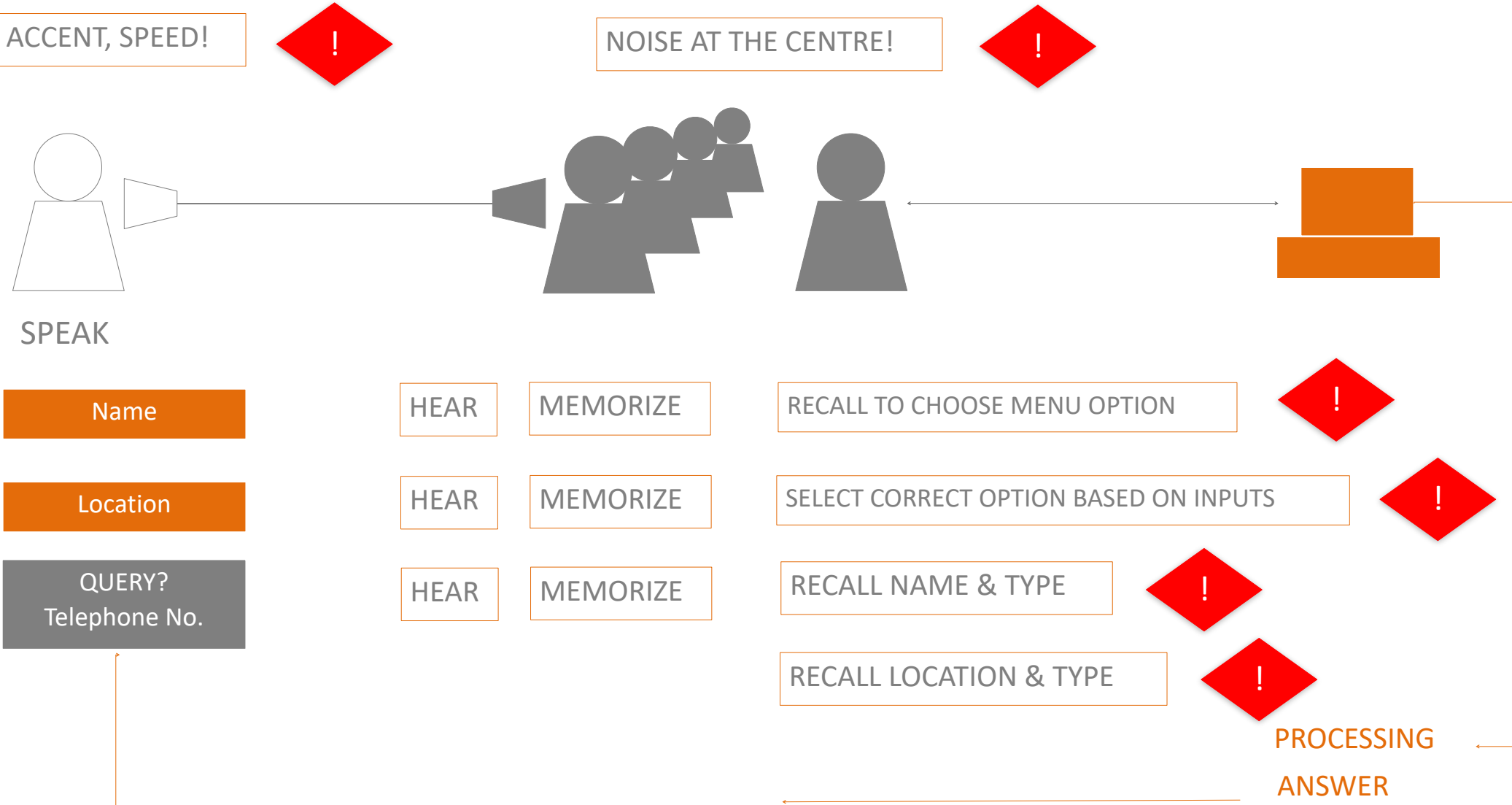
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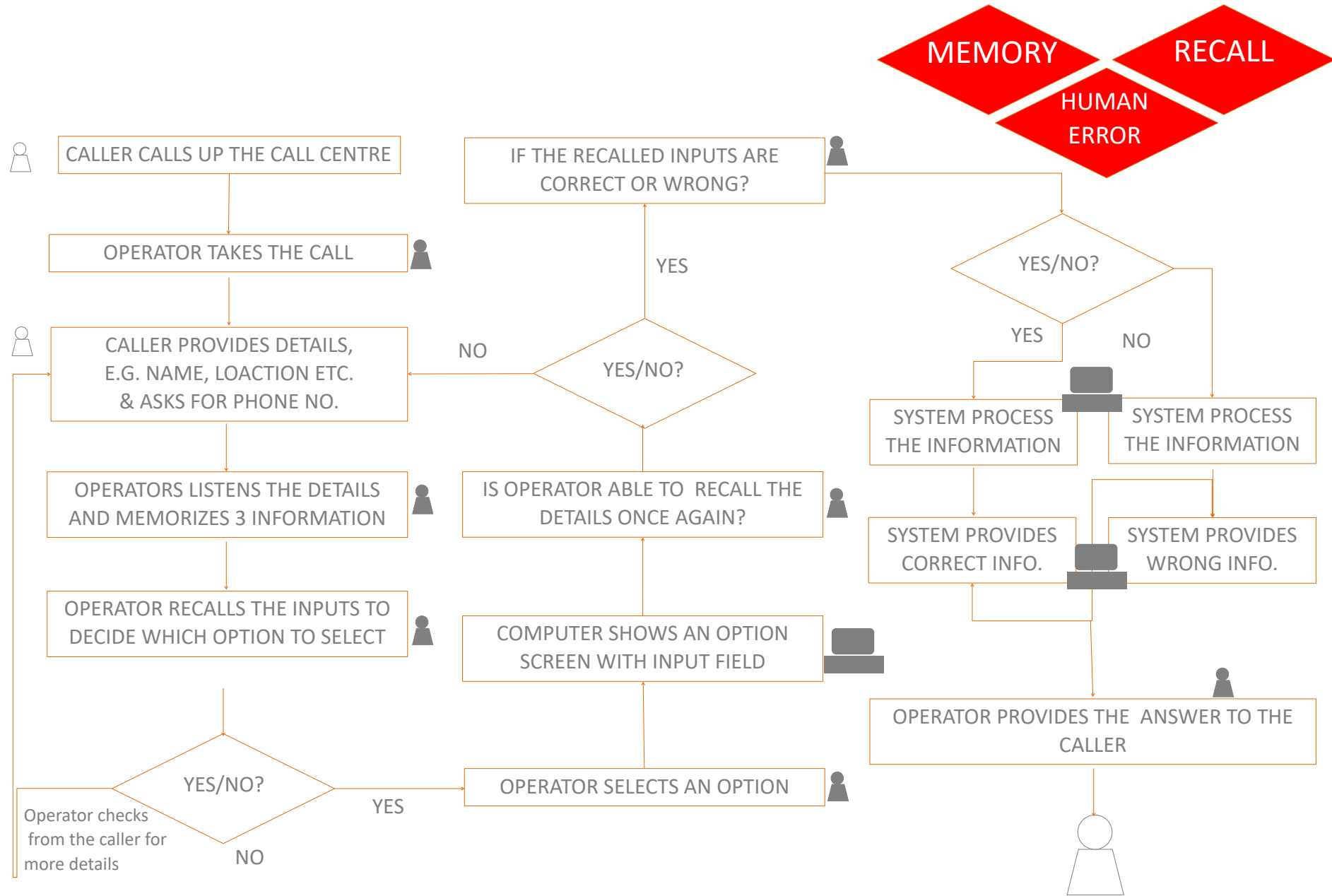
Distribution of Task between Human & Machine



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